A PROPOSAL TO DEVELOP SPMVV STUDENT SATELLITE – PAYLOAD OR SENSOR PART DESIGN DETAILS

Author names: M.Vani Ujwala; P.Vineela; Dr.V.Raja Rajeswari.

Department of ECE, School of Engineering and Technology (SOET), Sri Padmavathi Mahila Vishwavidyalayam (SPMVV), Tirupathi.

E-mail: mallembrnprathap@gmail.com, venelaponugoti@gmail.com

ABSTRACT

The payload and sensor component is the mission-specific "core" of a student CubeSat, designed to collect data for scientific, technological, or observational objectives while interfacing with the bus subsystems for power, data handling, and thermal control. Payloads are constrained by CubeSat size (e.g., 1U for simple sensors, 6U for advanced imagers), mass, and power limits, often using COTS optical cameras (VIS/NIR/IR), spectrometers, radiometers, or particle detectors. Earth observation payloads include multi-spectral imagers (e.g., 100m resolution in Bison Sat) for monitoring vegetation or disasters, requiring NOAA licensing for remote sensing. Communication payloads feature AIS receivers with deployable antennas for ship tracking, as in Tohoku University's 3U CubeSat. In-situ sensors measure space environment parameters like magnetic fields (e.g., ANDESITE's picosatellite mesh for auroral currents) or plasma (e.g., Prelude's electric field probes for earthquake precursors). Technology demonstrations include de-orbit sails (FREEDOM) or electrodynamic tethers (ALE-EDT) for debris mitigation. Biological experiments, like Bio Sentinel's yeast radiation studies, leverage microgravity. Development involves modular integration, HRM for deployments, and testing for vibration/shock/thermal extremes. Students must optimize resource allocation (e.g., data throughput for high-res images) and ensure non-interference, with examples like S-CUBE's meteor camera using gravity gradient booms for stabilization.